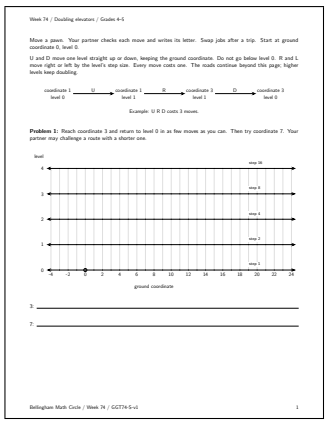
Week 74 / Doubling elevators / Page menu

For adults. Band slices print the pages whose range includes the band, with page and problem numbers unchanged.
K–1: no pages · Grades 2–3: no pages · Grades 4–5: whole packet


Week 74 / Doubling elevators / Grades 4–5

Move a pawn. Your partner checks each move and writes its letter. Swap jobs after a trip. Start at ground coordinate 0, level 0.

U and D move one level straight up or down, keeping the ground coordinate. Do not go below level 0. R and L move right or left by the level's step size. Every move costs one. The roads continue beyond this page; higher levels keep doubling.

coordinate 1 level 0 —U→ coordinate 1 level 1 —R→ coordinate 3 level 1 —D→ coordinate 3 level 0

Example: U R D costs 3 moves.

Problem 1: Reach coordinate 3 and return to level 0 in as few moves as you can. Then try coordinate 7. Your partner may challenge a route with a shorter one.

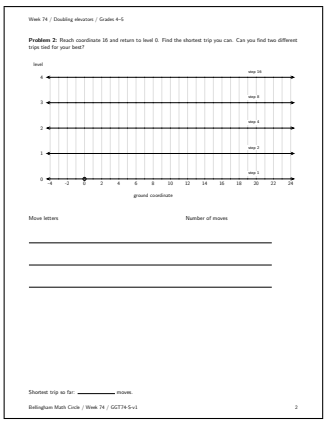
level
4
3
2
1
0
step 16
step 8
step 4
step 2
step 1
−4 −2 0 2 4 6 8 10 12 14 16 18 20 22 24
ground coordinate

3:

7:

Bellingham Math Circle / Week 74 / GGT74-S-v1 1

page 1 · 4–5 · P1


Week 74 / Doubling elevators / Grades 4–5

Problem 2: Reach coordinate 16 and return to level 0. Find the shortest trip you can. Can you find two different trips tied for your best?

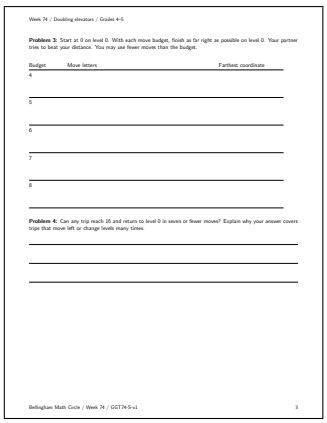
level
4
3
2
1
0
step 16
step 8
step 4
step 2
step 1
−4 −2 0 2 4 6 8 10 12 14 16 18 20 22 24
ground coordinate

Move letters Number of moves

Shortest trip so far: _____ moves.

Bellingham Math Circle / Week 74 / GGT74-S-v1 2

page 2 · 4–5 · P2


Week 74 / Doubling elevators / Grades 4–5

Problem 3: Start at 0 on level 0. With each move budget, finish as far right as possible on level 0. Your partner tries to beat your distance. You may use fewer moves than the budget.

Budget Move letters Farthest coordinate

4

5

6

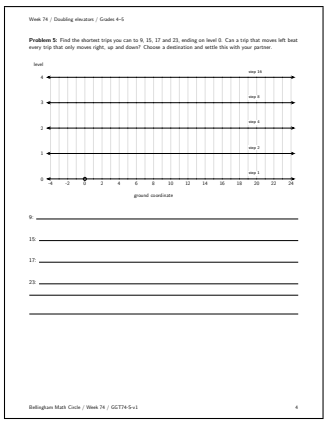
7

8

Problem 4: Can any trip reach 16 and return to level 0 in seven or fewer moves? Explain why your answer covers trips that move left or change levels many times.

Bellingham Math Circle / Week 74 / GGT74-S-v1 3

page 3 · 4–5 · P3–4


Week 74 / Doubling elevators / Grades 4–5

Problem 5: Find the shortest trips you can to 9, 15, 17 and 23, ending on level 0. Can a trip that moves left beat every trip that only moves right, up and down? Choose a destination and settle this with your partner.

level
4
3
2
1
0
step 16
step 8
step 4
step 2
step 1
−4 −2 0 2 4 6 8 10 12 14 16 18 20 22 24
ground coordinate

9:

15:

17:

23:

Bellingham Math Circle / Week 74 / GGT74-S-v1 4

page 4 · 4–5 · P5